

12/20

classmate
Date _____
Page _____

- 1) 1) write a program to print the multiplication table of a given number using a for loop

```
→ for i in range(1,6):
    print(f"::: Table of {i} :::")
    for j in range(1,11):
        print(i, "x", j, "=", i*j)
    print()
```

o/p:-

1x1=1	2x1=2	3x1=3	4	5
1x2=2	2x2=4	3x2=6	—	—
1x3=3	2x3=6	3x3=9	—	—
1x4=4	2x4=8	3x4=12	—	—
1x5=5	2x5=10	3x5=15	—	—
1x6=6	2x6=12	3x6=18	—	—
1x7=7	2x7=14	3x7=21	—	—
1x8=8	2x8=16	3x8=24	—	—
1x9=9	2x9=18	3x9=27	—	—
1x10=10	2x10=20	3x10=30	—	—

- 2) write a program to print all prime numbers between 1 & 100 using a for loop

```
→ for num in range(1,100):
    prime = True
    for j in range(1, num):
        if num % j == 0:
            prime = False
    if prime:
        print(num)
```

- 3) num = int(input)

2) 1) write a program to reverse a given number using a while loop.

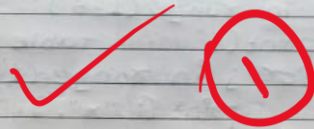
i = 10 ✓

j = 0

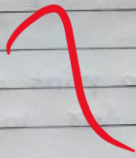
while i > j:

print(i)

i = i - 1



2) write a program to check whether a given number is a palindrome using a while loop



3) 1) write a function in python? explain its advantage:

→ function is a reusable block of code that perform a specific task ✓

Advantage :-

- Avoid code repetition ✓
- make code readable ✓
- Improve modularity ✓
- easier debugging ✓
- Resuability ✓

2) write a function that takes a list as input & return the largest element

→ num1 = int(input("Enter a first number:"))

num2 = int(input("Enter a 2nd number:"))

num3 = int(input("Enter a 3rd number:"))

if num1 > num2 and num1 > num3:

print(f"{num1} is greater than {num2}")

elif num2 > num1 and num2 > num3:

print(f"{num2} is greater than {num1}")

elif num3 > num1 and num3 > num2:

print(f"{num3} is greater than {num1}")

1) write a function to calculate the factorial of a number

→ def factorial(n):

if n == 1:

return 1

return n * factorial(n-1)

factorial(4)

5) userdefined fun	built-in fun
- It is created by the user/ prog	it is already available in python
- It is defined using def keyword	It is already defined keyword ✓

3) ✓ (X)

Q.4) 1) What is object-oriented programming (oop)?
→ oop is a programming paradigm. A programming paradigm is a fundamental style or way of thinking about how you structure & write your code.

2) Difference between class & object with ex:

class	object
- A class is a blueprint (template) used to create objects.	An instance is a real object created from a class
- class student: • class Name: student	obj = student("Gauri", 13) Here, • obj is the instance (object) of the student class

3) Four pillars of oop?

→ 1) Inheritance:

- a) single inheritance ✓
- b) multiple inheritance ✓
- c) multilevel inheritance ✓
- d) Hierarchical inheritance ✓
- e) Hybrid inheritance ✓

2) Encapsulation :- a) public b) protected c) private

3) Polymorphism :- a) method overriding b) method overloading

4) Data Abstraction ✓

4) What is inheritance? explain types.

→ Inheritance allows child class & parent class. One class inherits from another class. One class acquires the properties of another class.

- PRICE PAID _____
DATE / /
- a) single inheritance :- single inheritance a child inherits from only one parent class ✓
 - b) multiple inheritance :- multiple inheritance a child class inherits from more than one parent class
 - c) multilevel inheritance :- multilevel inheritance inheritance happens in levels ✓
 - d) Hierarchical inheritance :- Hierarchical inheritance multiple child classes inherits from a single parent class ✓
 - e) Hybrid inheritance :- is a combination of two or more inheritance type. ✓

- Polymorphism & explain method
- Polymorphism means many forms.
one function or method but multiple behaviors
- 1) method overloading :- python does not support tradition overloading it is using default argument ✓
- 2) method overriding :- child class provides its own implementation of parent method ✓

- encapsulation & its advantages &
- encapsulation is achieved using Access modifiers: Public, Protected, Private.
- Advantage :- 1) it is secure 2) data is protected 3) it is used Public, Protected, Private. ✓

Abstraction & implemented in python.
data Abstraction means hiding internal implementation details. ✓

8) Difference betⁿ Abstraction & encapsulation

Abstraction	encapsulation
1) Implementation hiding ✓	Data hiding ✓
2) How data is Protected	what functionality is provided ✓
3) Access modifiers	Abstraction